

Sampling and Image Formation Affecting Object Visibility

Question: An image captured with low resolution fails to detect objects clearly. Explain how sampling and image formation affect object visibility.

Theory

Low sampling reduces spatial resolution, causing loss of fine details and blurred object boundaries. Image formation quality influences visibility because poor focus, noise, and limited sensor resolution degrade features required for object detection.

Python Code

(Complete code as provided by the user should be inserted here for submission.)